

Platform Overview

VagaShield is an enterprise-grade, on-premise security operations platform engineered for the unique challenges of IT/OT convergence. It consolidates Security Information & Event Management (SIEM), Asset Intelligence, Network Visibility, and Compliance Management into a single, unified console. Designed for mission-critical environments, VagaShield provides deep visibility and threat detection across air-gapped infrastructure without reliance on cloud connectivity.

Key Positioning

- **Unified SIEM & SOC:** Centralized log management and threat detection.
- **Asset Intelligence:** Automated discovery of IT and OT/IoT devices.
- **Network Visibility:** Real-time topology mapping and link monitoring.
- **Compliance:** Integrated reporting for regulatory standards.

Deployment Options

- **On-premise Appliance:** Dedicated hardware solution.
- **Virtual Machine:** VMware ESXi, Hyper-V, KVM.
- **Bare-metal:** Direct installation on compliant hardware.



System Architecture

VagaShield leverages a proprietary modern microservices architecture designed for high availability, scalability, and security in isolated environments.

- Modern Microservices Architecture
- Advanced log ingestion and processing pipeline
- Enterprise-grade security monitoring core
- Interactive visualization and reporting suite
- Automated network discovery engine
- High-performance distributed data engine
- Real-time search and analytics engine
- Scalable caching and message queuing system
- Comprehensive infrastructure monitoring
- Full air-gapped and offline operation support

Key Feature Sets

1. SIEM & Security Operations Center (SOC)

- Unified SOC Dashboard with customizable widgets
- Event Explorer with high-speed full-text search
- Integrated Incident & Case Management workflow
- User & Entity Behavior Analytics (UEBA)
- Custom Detection Rules (SIGMA-compatible)
- Advanced Multi-source Correlation Engine
- Threat Intelligence Integration (STIX/TAXII)
- MITRE ATT&CK Navigator mapping

2. Asset Intelligence & Discovery

Gain complete visibility into every device connected to your network, from servers to industrial controllers.



Figure 2: Comprehensive Asset Inventory and Risk Scoring Interface.

- Comprehensive IT/OT Asset Database (CMDB)
- Active & Passive Network Scanning capabilities
- Advanced Device Fingerprinting & OS Classification
- Specialized IoT/OT Protocol Support (Modbus, OPC-UA, BACnet)

3. Network Visibility & Topology

Visualize your infrastructure with automated mapping to understand relationships and dependencies.

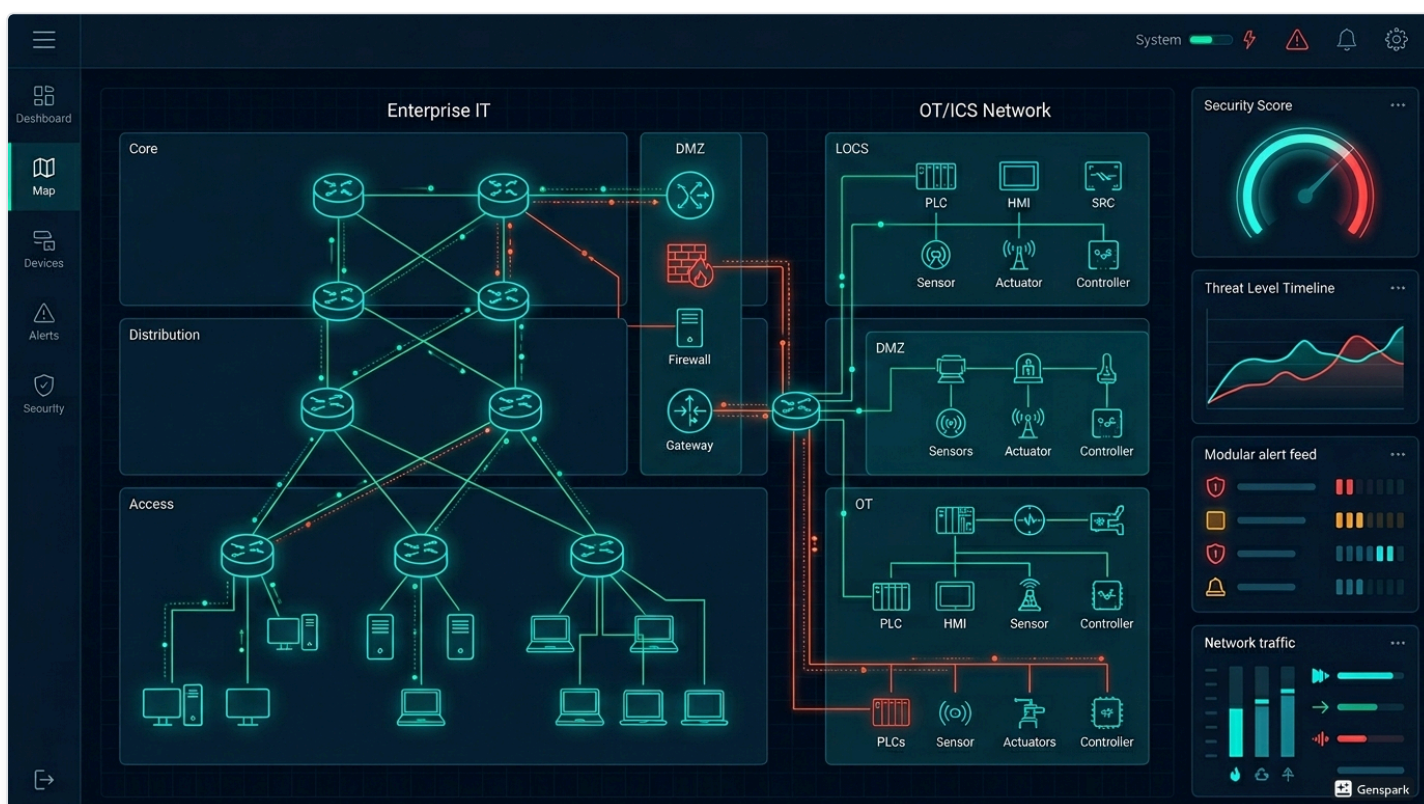


Figure 3: Auto-generated L2/L3 Network Topology Map.

- Auto-generated Layer 2 / Layer 3 Topology Maps
- Real-time Link Status Monitoring & Bandwidth Usage
- Automated Network Device Configuration Backup & Restore

4. Security & Vulnerability Management

- Unified Risk Scoring per asset
- Rogue Device Detection & Auto-Quarantine (MAC-based)
- CVE-mapped Vulnerability Scanning
- Configuration Assessment

5. Reporting & Analytics

- Custom Report Builder (Export to PDF, CSV, JSON)
- Automated Scheduled Report Distribution (Email/Share)
- Rich Widget Library for Executive Dashboards
- Regulatory Compliance Templates

6. Automation & Administration

- SOAR capabilities with Visual Playbook Builder
- Bi-directional Integration Hub (API/Webhooks)
- Role-Based Access Control (RBAC) & Multi-Tenancy
- Multi-Factor Authentication (MFA) & SSO
- OEM White-label Support for MSPs

System Requirements

Component	Minimum Specification	Recommended Specification
CPU	8 Cores	16+ Cores
RAM	16 GB	32+ GB
Storage	500 GB SSD	2+ TB NVMe
OS	Ubuntu 20.04+ LTS	Ubuntu 22.04 LTS
Network	1 Gbps Ethernet	10 Gbps Ethernet

Supported Log Sources

Syslog (UDP/TCP/TLS)
Windows Event Logs
Linux System Logs
Firewall Logs
Network Devices
Web Server Logs
Database Audit Logs
Cloud Services (AWS/Azure)
EDR Telemetry
Custom Applications

Compliance & Standards

PCI-DSS v4.0
ISO 27001:2022
NIST CSF
SOC 2 Type II
CIS Benchmarks
GDPR
Thai PDPA